$$\ldots \underbrace{D\ C\ B\ A}$$

Last 4 digits of your student ID.

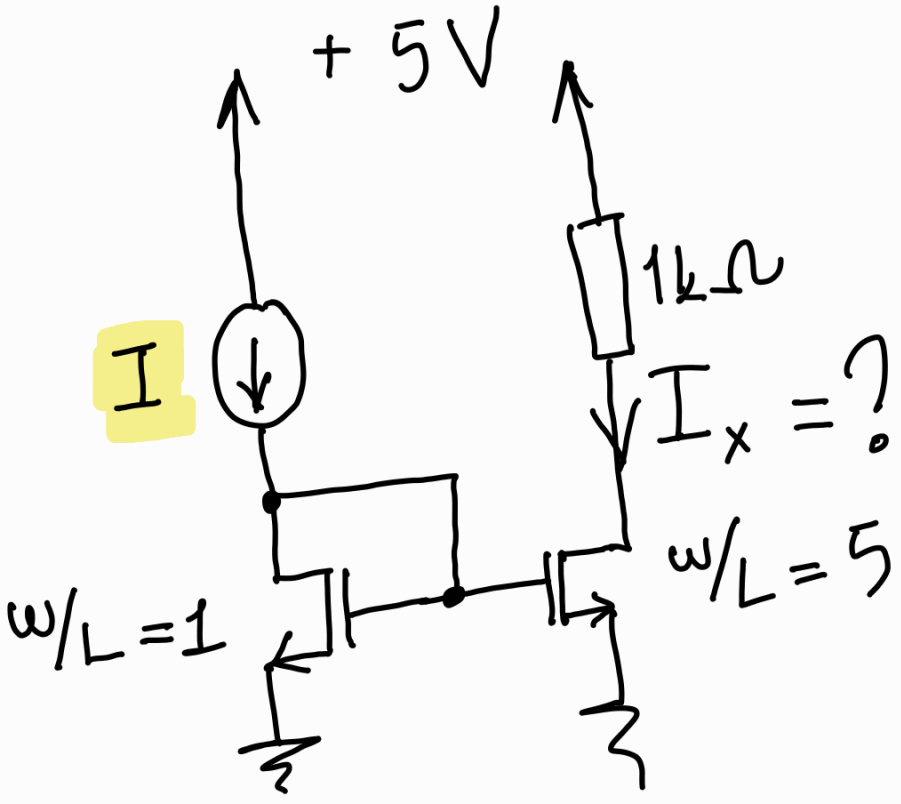
$$I = (A+B+2)\,\mu A$$

$$R = \left(\frac{A+C+5}{50}\right) k\Omega$$

$$V_S = (A+D+5)\,V$$

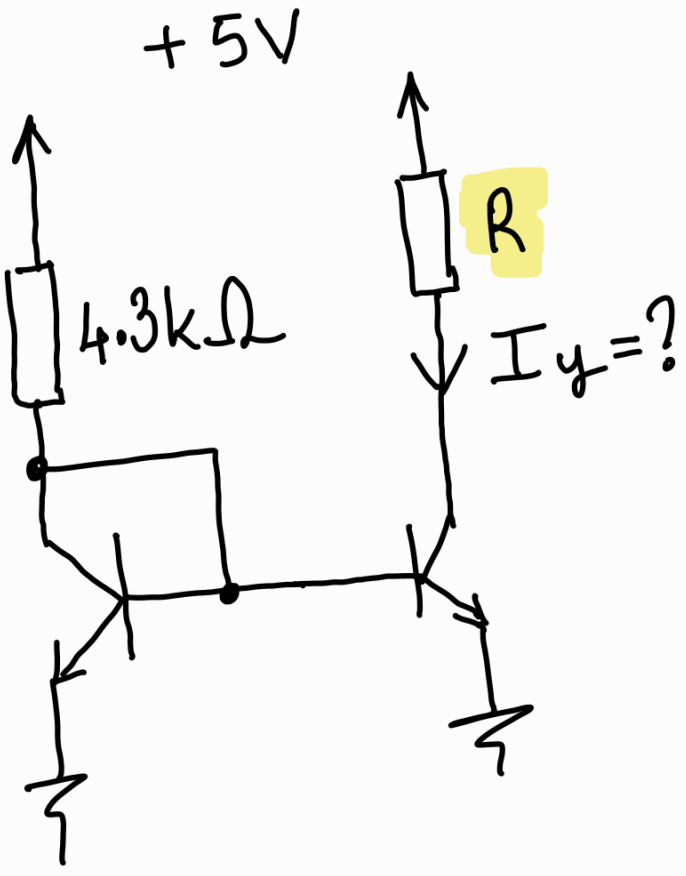
$$g_m = (B+C+5)\ mA/V$$

Q1. . . . D C B A $I = (A+B+2)\,\mu A$



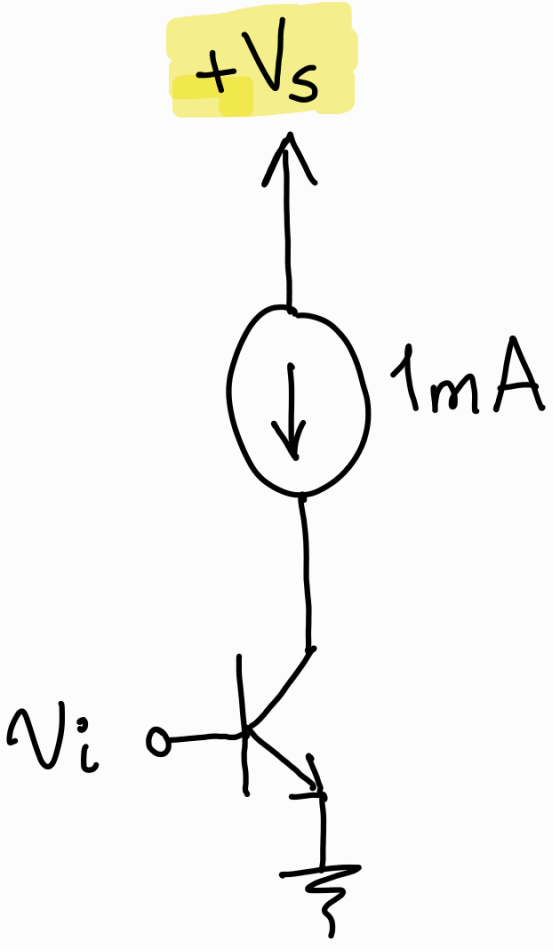
$I_x =$ ____ μA

Q2. . . . D C B A $R = \left(\frac{A+C+5}{50}\right) k\Omega$

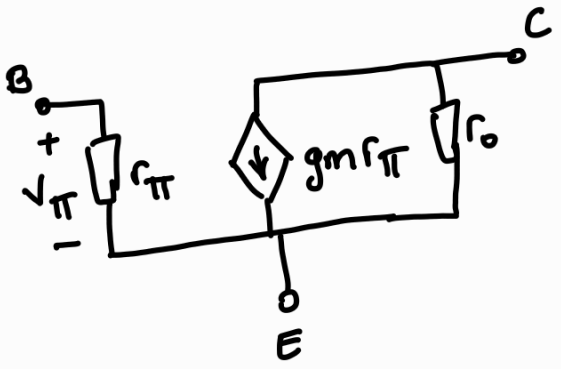


I_y = [ ] mA

Q3. . . . D C B A $V_S = (A+D+5)\,V$



Hybrid π-model for BJT

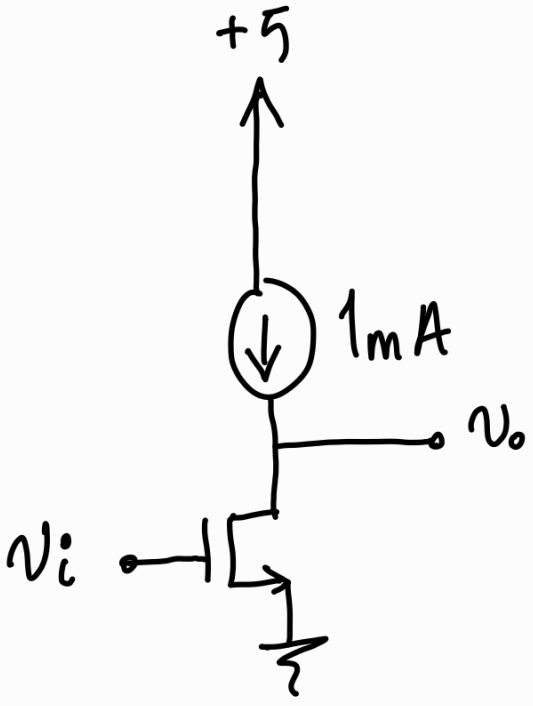


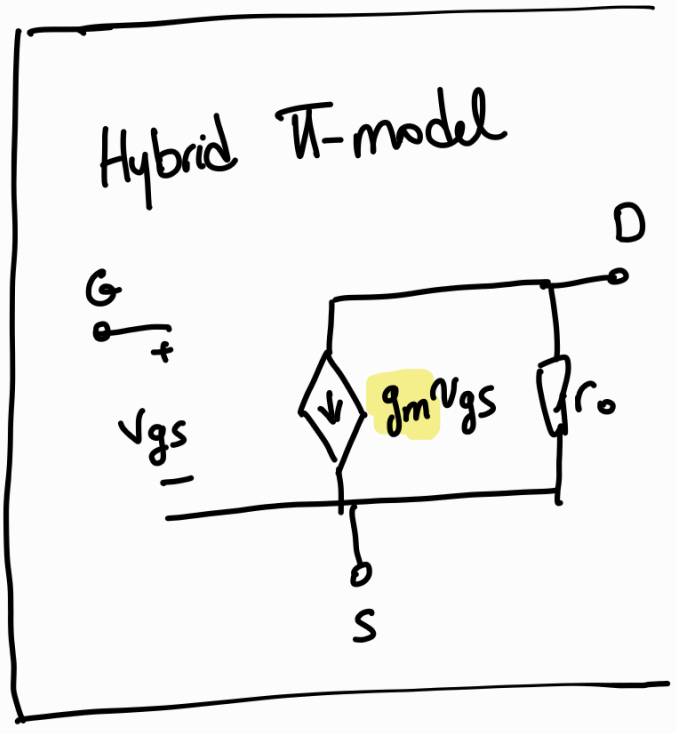
$$g_m = \frac{I_C}{V_T}$$

V_T @ 300°K = 0.025 V

$g_m =$ _____ mA/V

Q4. . . . D C B A $g_m = (B+C+5)$ mA/V

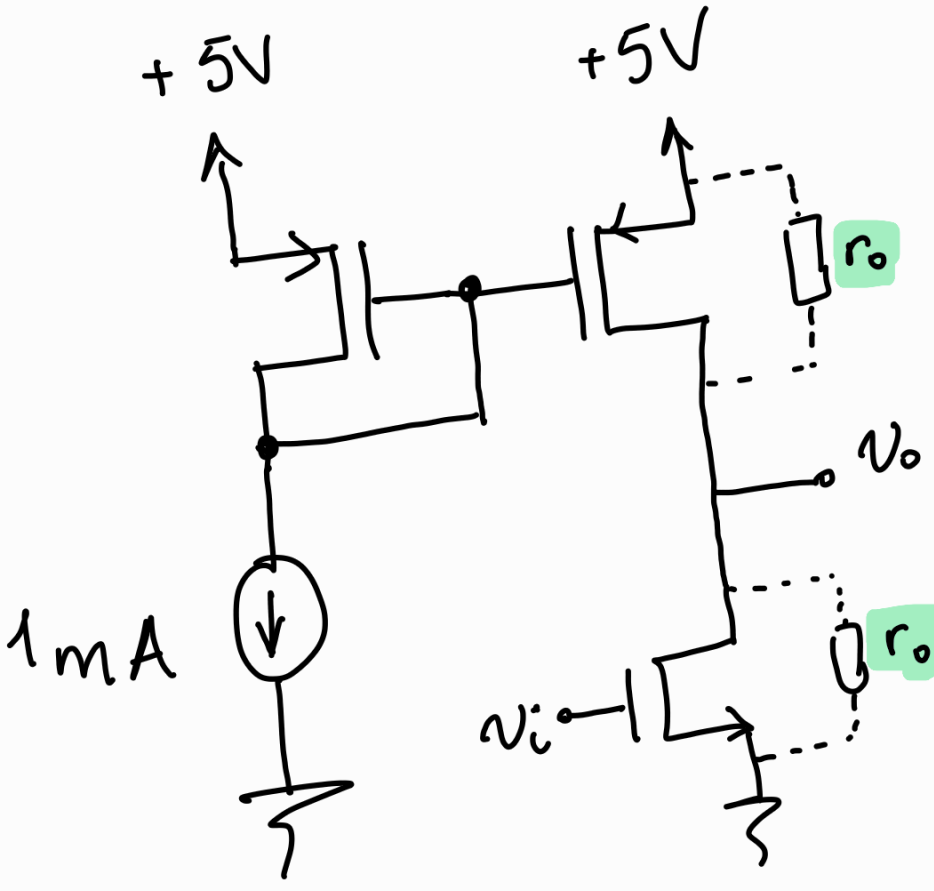




$r_o = 20k\Omega$

$A_o =$ V/V

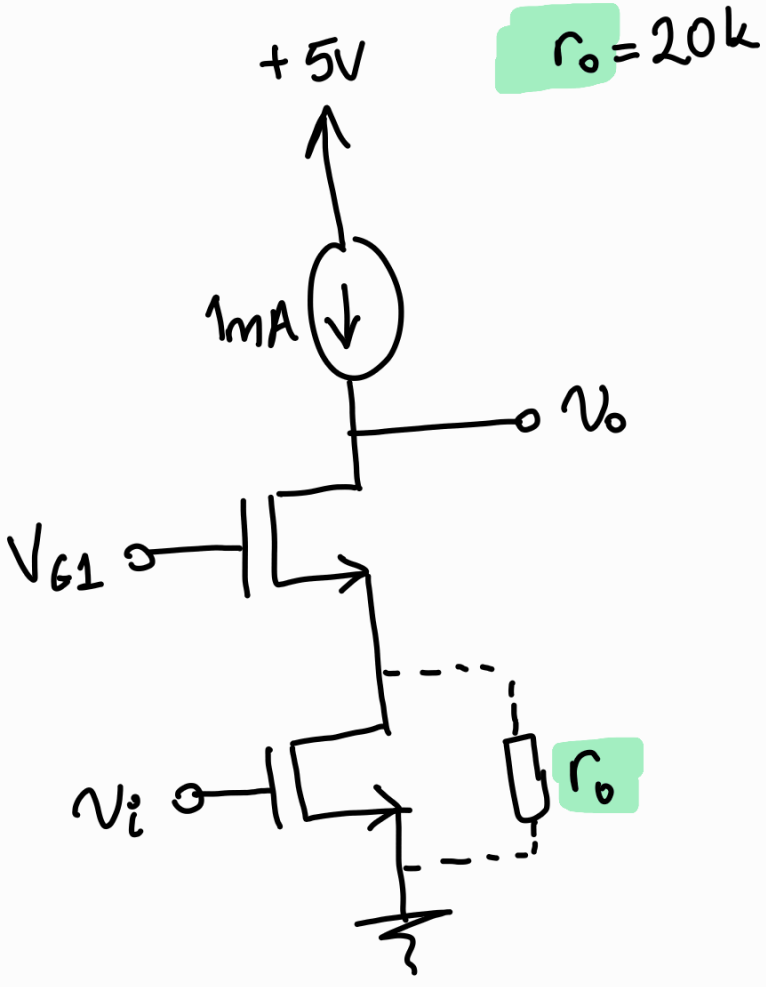
05. . . . D C B A $g_m = (B+C+5)$ mA/V

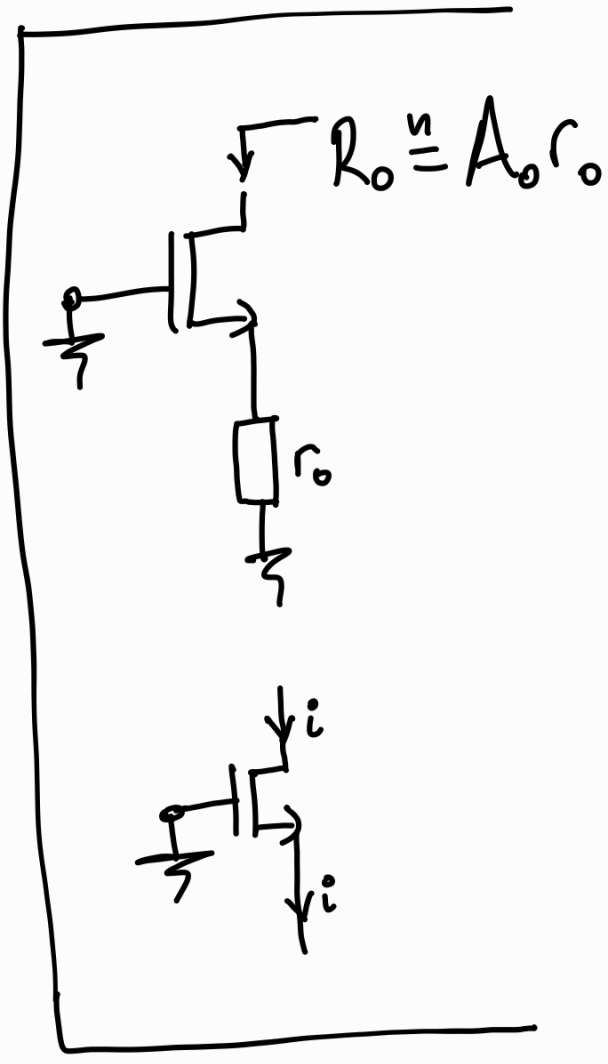


$r_o = 20k$

$A_{v_0} = v_o/v_i =$ V/V

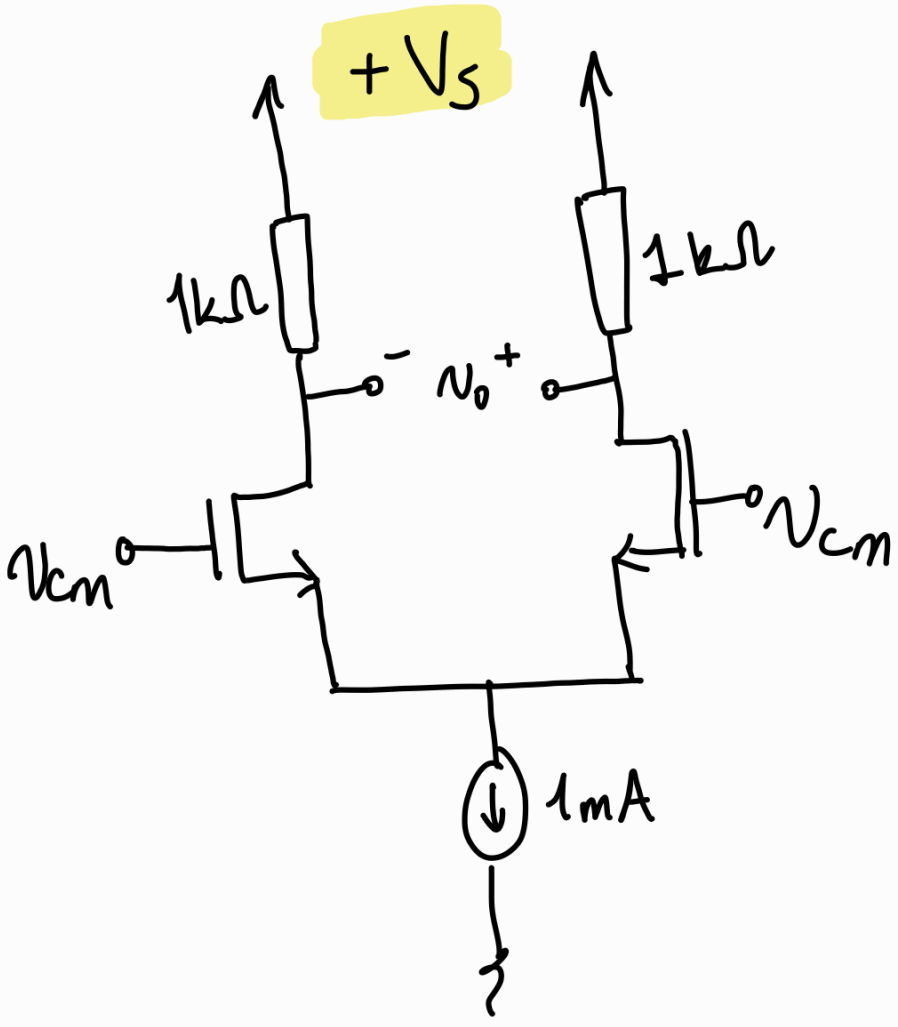
Q6. . . . D C B A $g_m = (B+C+5)$ mA/V





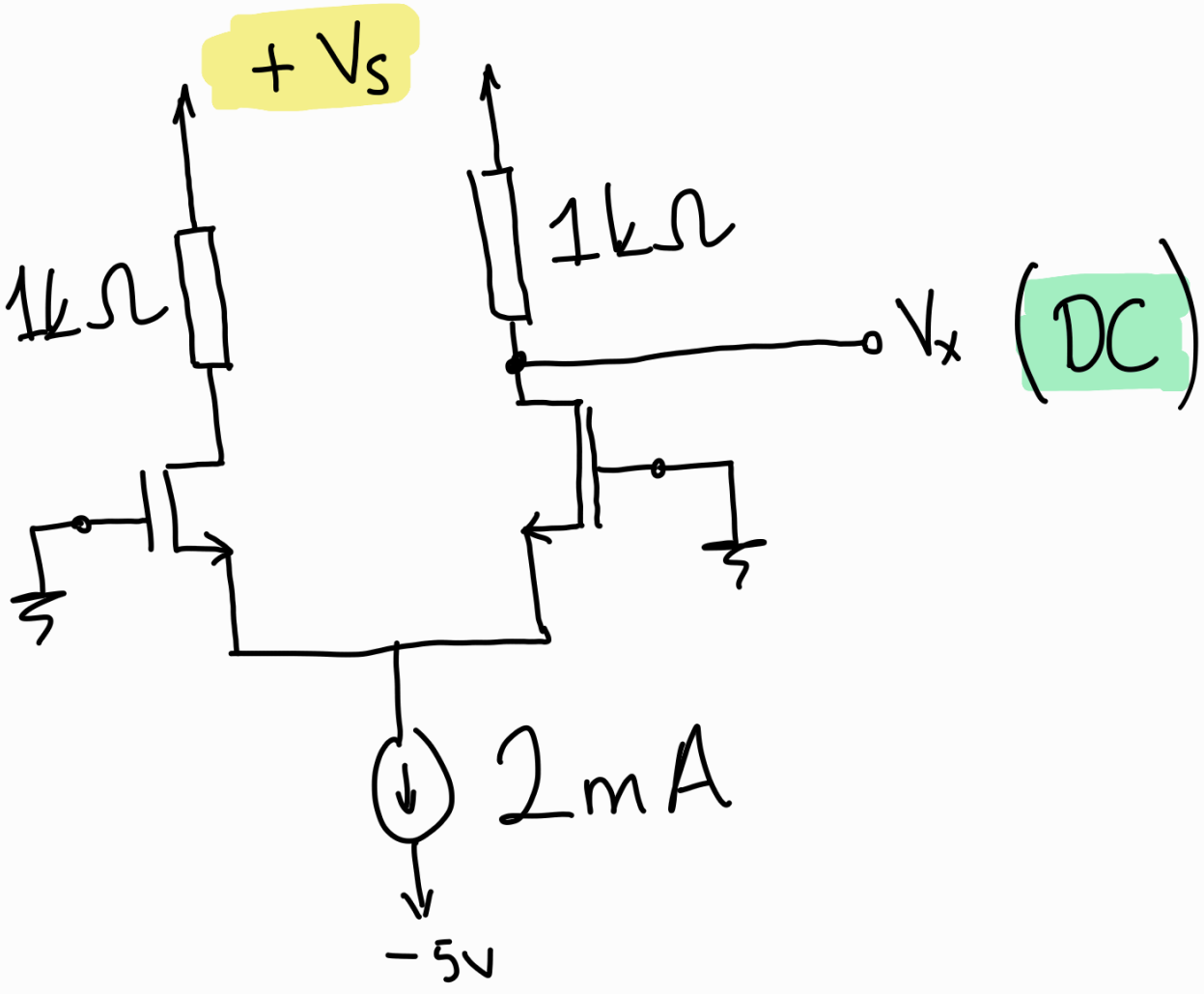
$$A_v = \frac{v_o}{v_i} = \quad V/V$$

Q7. . . . D C B A $V_S = (A+D+5)\,V$



$$A_{cm} = \frac{v_o}{v_{cm}} = \qquad V/V$$

08. . . . D C B A $V_S = (A+D+5)\,V$

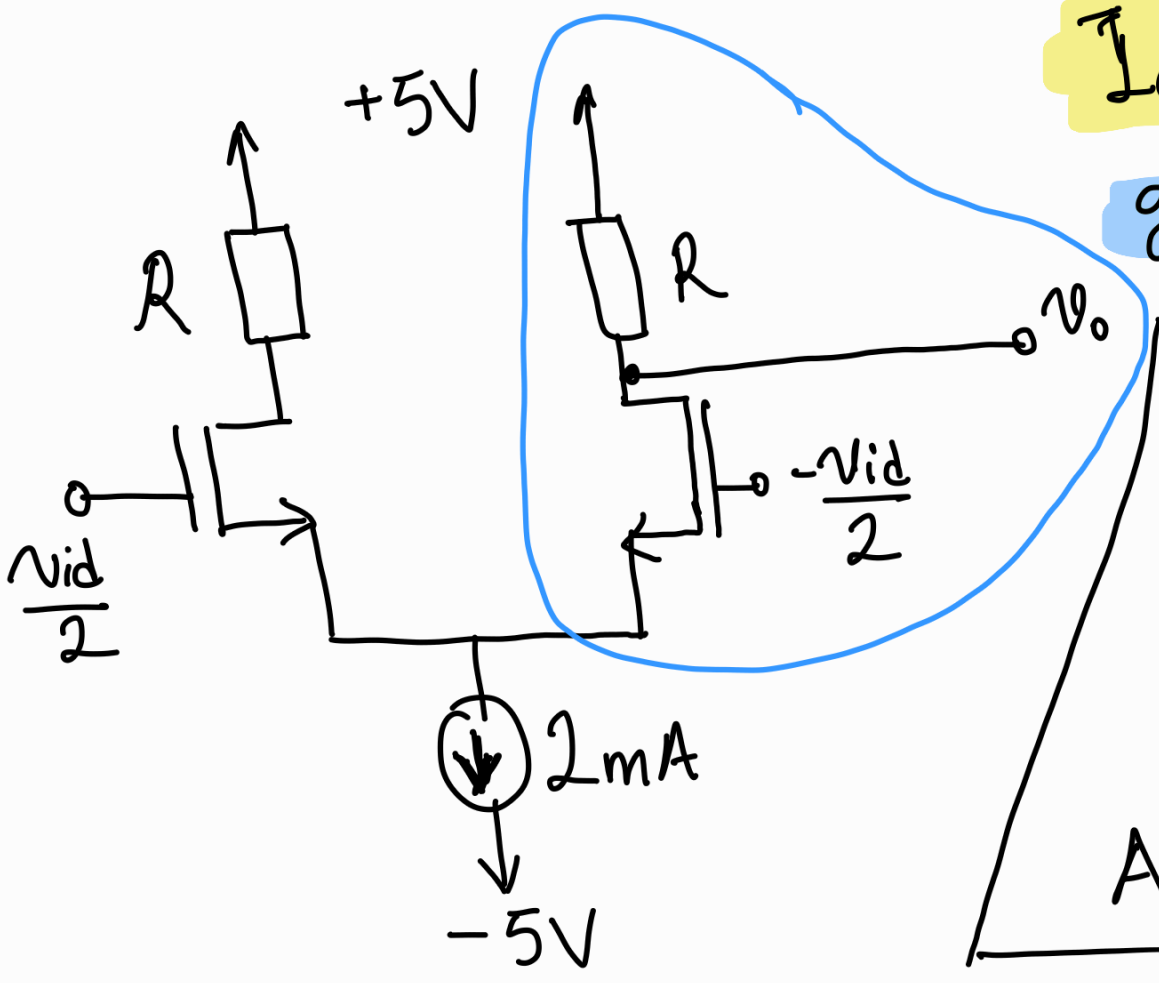


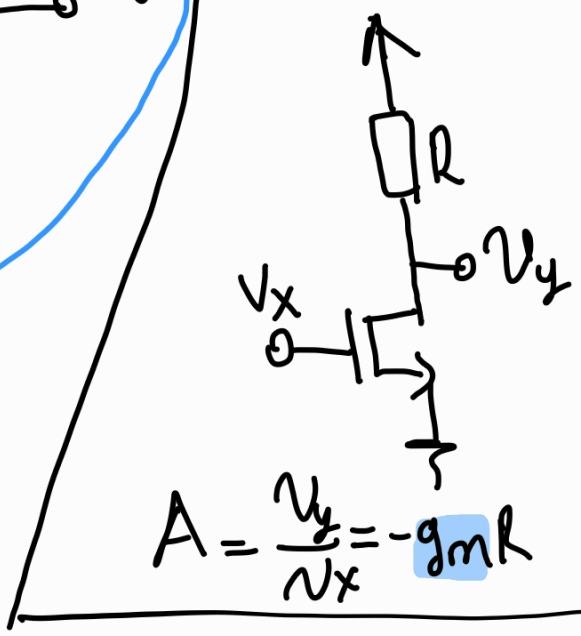
$V_x =$ V

Q9. . . . D C B A $R = \left(\frac{A+C+5}{50}\right) k\Omega$

Ignore r_o

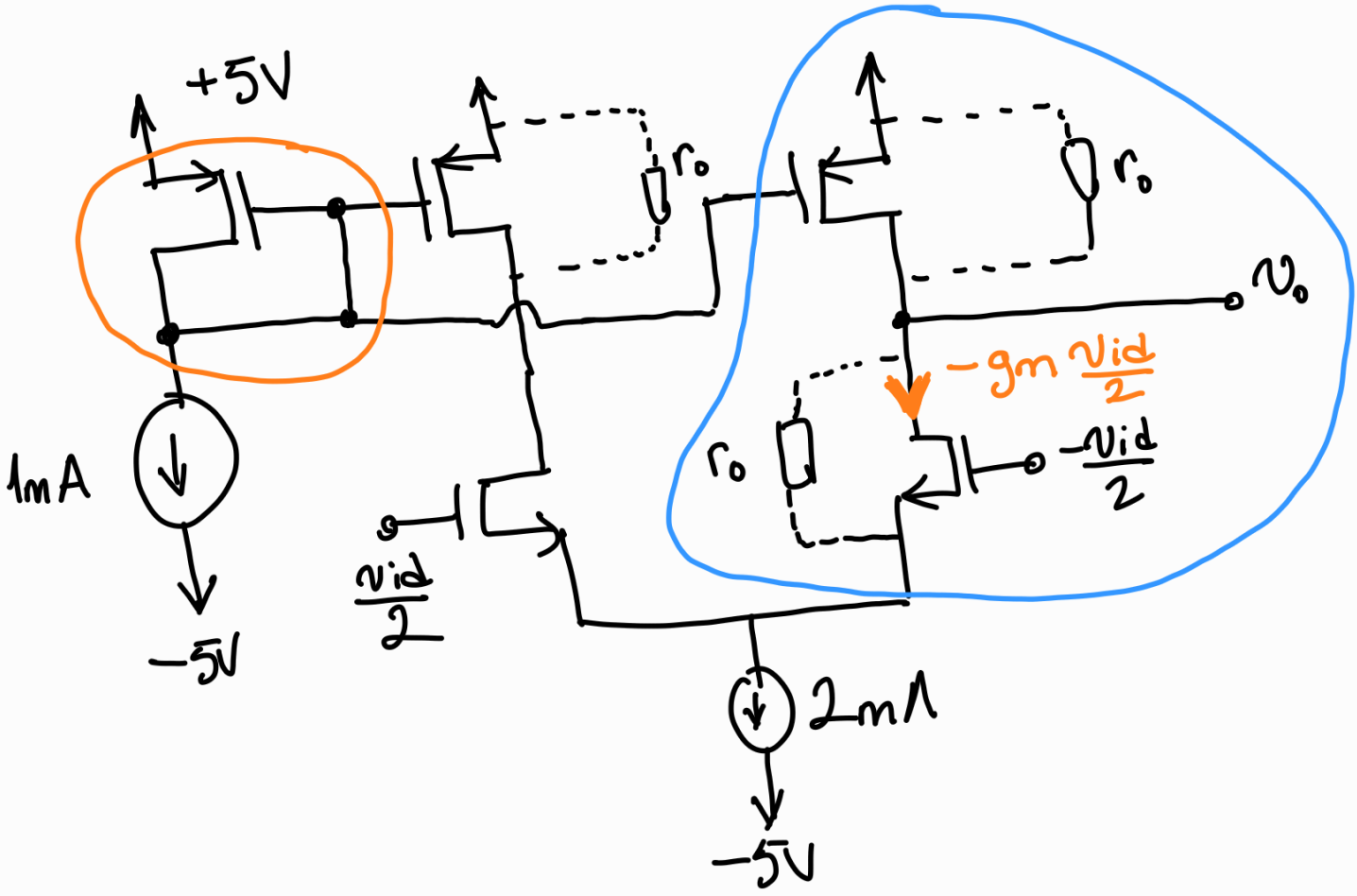
$g_m = 5 mA/V$





$A_d = v_o / v_{id} =$ V/V

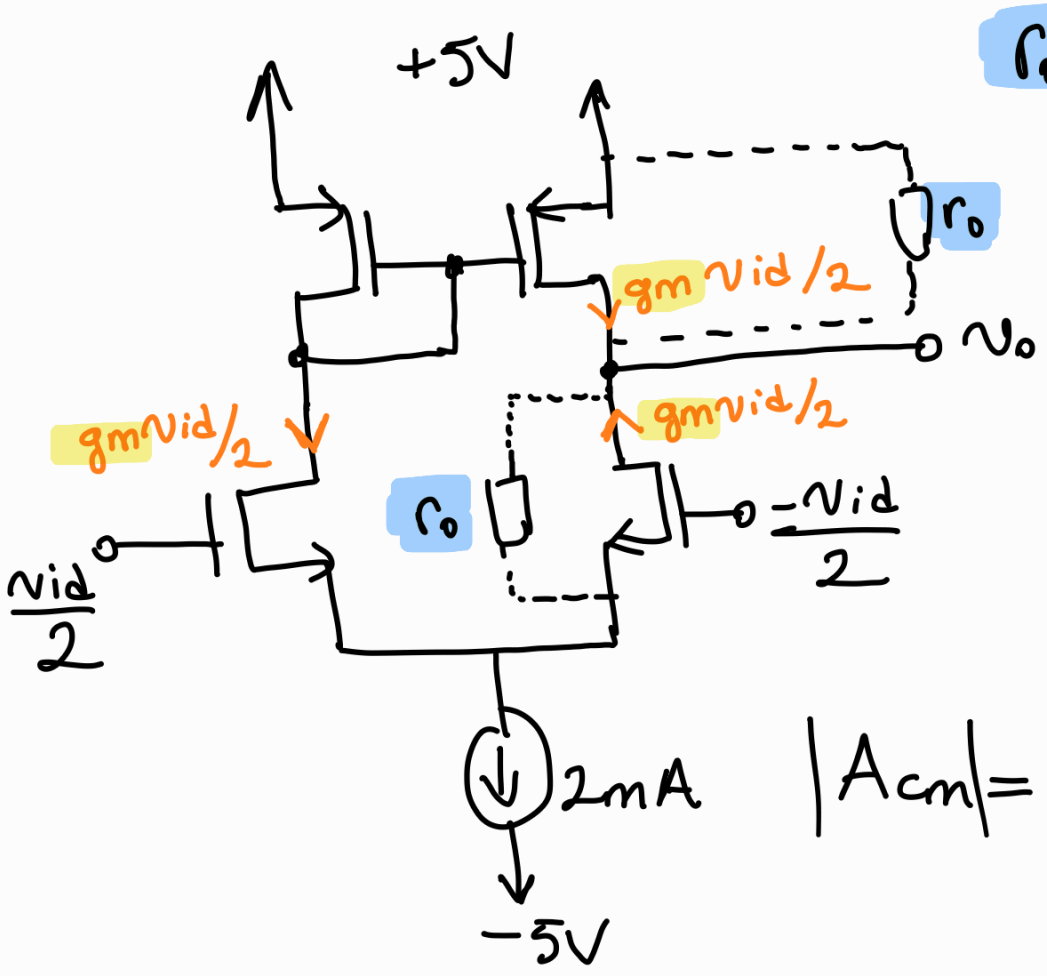
Q10. . . . D C B A $g_m = (B+C+5)$ mA/V



$$A_d = \frac{v_o}{v_{id}} = \quad V/V$$

Q11. . . . D C B A $g_m = (B+C+5)$ mA/V

$r_o = 20k$



$|A_{cm}| = 0.01$ V/V

CMRR =